

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INTERACTIVE TREE SIMULATOR

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 1 – Interactive Tree Simulator
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	<i>Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.</i>		
Student Name:	M.Deepika	Reg. No.:	192372296
Section / Batch:	2023	Date:	29/07/2026

Code of Conduct

I _____M.DEEPIKA-192372296_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____deepika.m____

Instructions

- This assessment tool contains 5 Interactive Tree Simulator questions, Total: 100 Marks.
- Students can use any online/offline Interactive Tree Simulator. Demonstrate insertion, deletion, searching, and traversal operations wherever applicable.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
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Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	1	20
Tree Traversals	Traversal types, In order, Post order and Pre order	2	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	3	20
B Tree, 2-3 tree, 2-3-4 tree	Properties, structures and Operations	4	20
Red Black Tree	Properties, Example and Operations	5	20
GRAND TOTAL:			100 Marks

Annexure A – Interactive Tree Simulator

- Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80
Perform:
 - Inorder Traversal
 - Search for 60
 - Delete 30
 - Display the final BST.
- Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80. Find all the traversals.
 - Inorder
 - Preorder
 - Postorder
- Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.
- Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75
- Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Correct Tree Construction	20	Tree is constructed correctly with all operations performed accurately	Minor errors in construction	Some operations incorrect	Tree construction mostly incorrect
Understanding of Tree Operations	20	Clearly explains insertion, deletion, search and balancing operations	Explains most operations correctly	Limited understanding	Unable to explain operations

Analysis of Tree Behaviour	30	Correctly analyses balancing, rotations, node split/merge, splaying	Good analysis with minor omissions	Partial analysis	Poor or incorrect analysis
Presentation	20	Well-organized report with accurate observations and conclusion	Good report with minor issues	Basic report with limited observations	Incomplete report
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80 Perform:

a. Inorder Traversal

The screenshot shows the 'Tree Traversals' web application interface. The browser tabs include 'CSA0305 - Slot A-Course Links', 'CSA0305-Data-Structure-/ASSE...', 'slot-a-csa0305-assessments/co...', and 'Data Structure Visualizer'. The address bar shows 'tree-graph-algorithm-visualizer.vercel.app/treetraversals'. The application has a dark theme. On the left, the 'Controls' panel includes an 'Input' field with the text '50, 30, 70, 20, 40, 60, 80', a 'Traversal Order' section with 'Pre', 'In', and 'Post' buttons (where 'In' is selected), and buttons for 'Generate Default', 'Visualize', and 'Start Simulation'. The main area displays a binary search tree diagram with root 50, left child 30, right child 70, and leaf nodes 20, 40, 60, and 80. Above the tree are seven empty boxes for the traversal result. The bottom of the browser shows a taskbar with a search bar and various icons, and a system tray indicating 36°C and 01:53 PM on 18-08-2026.

This screenshot shows the same 'Tree Traversals' application after performing an inorder traversal. The 'Input' field and 'Traversal Order' (with 'In' selected) remain the same. The 'Visualize' button has been clicked. The main area now displays a message box with the text 'Traversal result: [20, 30, 40, 50, 60, 70, 80]' and an 'OKAY' button. The binary search tree diagram is still visible below the message box. The browser tabs and address bar are identical to the previous screenshot. The system tray at the bottom shows the same time and date, but the temperature is now 36°C and the status is 'Mostly sunny'.

b.Search for 60

CSA0305 - Slot A-Course Links

CSA0503-Data-Structure-/ASSE

slot-a-csa0305-assessments/co

Data Structure Visualizer

Ask Gemini

tree-graph-algorithm-visualizer.vercel.app/depthfirstsearch

Back to Home

Depth First Search

Input

Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Target Value

60

Generate Default

Visualize

Start Simulation

50

30

70

20

40

60

80

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36°C Mostly sunny

Search

ENG IN

01:58 PM 18-08-2026

tree-graph-algorithm-visualizer.vercel.app/depthfirstsearch

Back to Home

Depth First Search

Input

Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Target Value

60

Generate Default

Visualize

Start Simulation

50

Found target 60 at node root.right.left.

OKAY

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36°C Mostly sunny

Search

ENG IN

01:59 PM 18-08-2026

C. Delete 30

Insert(node.right, key)

Enter a number: 31

INSERT DELETE CLEAR

```
graph TD; 50((50)) --> 40((40)); 50 --> 70((70)); 40 --> 20((20)); 70 --> 60((60)); 70 --> 80((80));
```

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d.Display the final BST.

CSA0305 - Slot A-Course Link x CSA0503-Data-Structure-/AS x slot-a-csa0305-assessments/ Data Structure Visualizer AVL Tree Visualizer | Self-Bal

tree-graph-algorithm-visualizer.vercel.app/binarytree

Binary Tree

Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Visualize

Generate Python Code

```
graph TD; 50((50)) --> 30((30)); 50 --> 70((70)); 30 --> 20((20)); 30 --> 40((40)); 70 --> 60((60)); 70 --> 80((80));
```

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36°C Mostly sunny 02:14 PM 18-08-2026

2. Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80 Find all the traversals. a. Inorder

a. Inorder

tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80
Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order
Pre In Post
Generate Default
Visualize
Start Simulation

20 30 40 50 60 70 80

```
graph TD; 50 --> 30; 50 --> 70; 30 --> 20; 30 --> 40; 70 --> 60; 70 --> 80;
```

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tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80
Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order
Pre In Post
Generate Default
Visualize
Start Simulation

20 30 40 50 60 70 80

Traversal result: [20, 30, 40, 50, 60, 70, 80]

OKAY

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b. Preorder

tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order
Pre In Post

Generate Default

Visualize

Start Simulation

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tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order
Pre In Post

Generate Default

Visualize

Start Simulation

Traversal result: [50, 30, 20, 40, 70, 60, 80]

OKAY

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tree-graph-algorithm-visualizer.vercel.app/treetraversals

c. Postorder

tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order
Pre In Post

Generate Default

Visualize

Start Simulation

20 40 30 60 80 70 50

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Search

ENG IN

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tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order
Pre In Post

Generate Default

Visualize

Start Simulation

20 40 30 60 80 70 50

50

Traversal result: [20, 40, 30, 60, 80, 70, 50]

OKAY

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Search

ENG IN

02:24 PM 18-08-2026

3. Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.



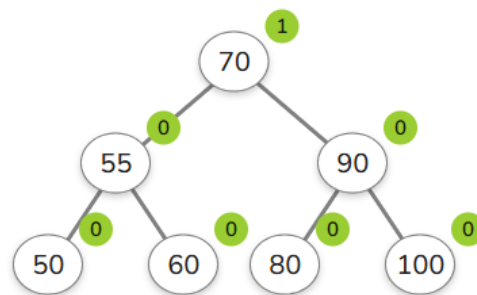
Enter a number: 40



INSERT

DELETE

CLEAR



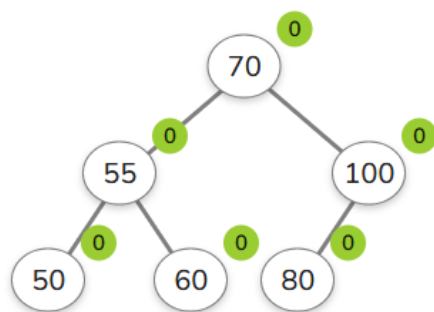
Enter a number: 7



INSERT

DELETE

CLEAR



4. Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75

SEE ALGORITHMS

Sorting
Graph
Finding MST
Data Structures
Advanced Trees
AVL Tree
Red-Black Tree
AVL Tree vs Red-Black Tree
Splay Tree
B-Tree
Other

written efficiently.

When inserting a new key, it is placed into the appropriate leaf node. If the node overflows by exceeding the maximum number of keys, it **splits** — the median key is pushed up to the parent, while the remaining keys form two child nodes. This process can propagate upward and may even create a new root. This splitting mechanism is essential for keeping the tree balanced, ensuring that all leaves always remain at the same depth as keys are redistributed.

Enter a number: 13

```
graph TD; 45 --> 25; 45 --> 65; 25 --> 15; 25 --> 35; 65 --> 55; 65 --> 75;
```

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Search

ENG IN

02:37 PM 18-08-2026

5. Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.

Enter a number: 26

```
graph TD; 100((100 B)) --> 80((80 B)); 100 --> 120((120 B)); 80 --> 60((60 R)); 80 --> 90((90 R)); 120 --> 110((110 R)); 120 --> 130((130 R));
```

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Search

ENG IN

02:42 PM 18-08-2026

